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Project Title:	NSCOR: Space Radiation and Intestinal Tumorigenesis: Risk Assessment and Counter Measure Development		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	http://www9.georgetown.edu/		
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Flight Program:			
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Task Description:	Ionizing radiation (IR) is a known risk factor for colorectal (CR) cancer. Considering the high spontaneous incidence of gastrointestinal (GI) cancer in the US and even higher incidence of pre-malignant lesions, such as colonic polyps, an even modest increase by radiation exposure could have a significant effect on health risk estimates for manned space flight. During space travel, astronauts are exposed to levels of radiation, which increases their risk for acute short-term deleterious effects, such as during a solar storm, and for long-term deleterious effects like cancer. Long duration space missions, like those at the International Space Station, lunar missions, or a mission to Mars, are of sufficient duration to allow for appreciable cumulative doses of cosmic radiation that could have profound health consequences. This project is important because there are no good estimates for intestinal, particularly CR, cancer by space radiation. Considering the increased CR cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors. We are employing APC mouse models that have been successfully used to demonstrate radiation-induced intestinal cancer. The APC models have inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. For our NSCOR study we are employing APC1638N model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa 1638. These mice develop very few adenomas in small as well as in large intestine but post-radiation tumor numbers shows marked increases. The CDX2P Apc flox/+ the other APC mouse model we will be employing has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequent adenocarcinomas in the colon. Data from these two mouse models will be combined with our previous studies on ApcMin/+ mice, a widely studied model for colorectal cancer to calculate risk estimates for NASA. The strategy is to compare results for space radiation, protons and HZE (56-Fe ions), on intestinal tumor development to that of low-LET (linear energy transfer) radiation (gamma rays). Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for humans exposed to space radiation. Many of the molecular events leading to intestinal tumors have been characterized. Normal human colonocytes with genetic manipulations representing major transitions in colon cancer initiation and progression will be used to monitor and compare molecular changes in three types of radiation-induced tumors. Isogenic strains that are normal and partially transformed will be exposed to high-energy protons and HZE nuclei and compared to gamma rays and tested for biological assays of malignant transformation. There will be the following three Projects addressing the impact of the following: 1) Space radiation on mouse models of intestinal tumorigenesis 2) Space radiation on neoplastic events in normal human colonocytes 3) Space radiation on GI stem cells (beginning in year 3)		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and the incidence of CR cancer in 2005 (latest year for which statistics are available) was 56.4 per 105 with a death rate of 21.0 per 105 or about 53,005 deaths per year. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of CR tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic and epigenetic processes affecting genome integrity and CR carcinogenesis.
Task Progress:	The overall plan is to apply the well-characterized APC mouse models of intestinal tumorigenesis for development of risks estimates for various radiation exposures during manned space flight. To undertake a comparative study of intestinal tumor risk in mice following exposure to different types of radiation, the relative toxicity of 56-Fe particles and protons to gamma-rays needed to be determined so that we can standardize with doses equitoxic to 0.5, 1.0 and 2 Gy of gamma-radiation in C57Bl/6J mice. The gamma-ray dose of 2 Gy has been shown to be effective in increasing intestinal tumorigenesis with 1 GeV 56-Fe and proton radiation. The rationale for this initial study is that acute toxicity for the planned radiation sources at the proposed energies are not available in mice. We submitted animal protocols to both our home institution and to Brookhaven National Laboratory (BNL) and now have approval for experiments at the NSRL. We have carried out two runs so far at the NSRL at BNL. Initial pilot experiments with a solar particle event (SPE) simulation at both NSRL 10A and 10B. The objective of these experiments is to determine if the incidence or progression of colon cancer is increased after exposure to a total body of 2 Gy protons. It will take about one year until we sacrifice the mice to determine if we observe an increase in the number of adenomas, adenoma-adenocarcinoma transitions, or the development of metastatic disease. As we already have breeding stock from our NASA grant, APC1638N/+ was used for quantitative and qualitative study on intestinal tumorigenesis at 5 Gy of gamma rays at GU, and to equitoxic doses of 1 GeV/n protons and 1 GeV/n 56-Fe ions at NSRL. Female and male mice were exposed at the age of 6-8 weeks and animals were sacrificed 150 days after irradiation. We observed that a striking difference exists in the susceptibility to gamma-induced intestinal tumors of female and male APC1638N mice. While for male mice we observed a ~15-fold increase in overall intestinal tumor multiplicity following exposure to 5 Gy of gamma rays, no significant increase was found in female mice compared to age-matched controls. Furthermore, we report here that in male mice 5 Gy of gamma rays and equitoxic doses of space radiation induce the same increase in overall intestinal tumor multiplicity. However, when the colon tumorigenesis was considered as an end-point, Fe ions induced the greatest effect on both tumor number and grade. In the course of year 1, we have also started our study on effects of 1 Gy of gamma rays and equitoxic doses of 1 GeV/n protons and 1 GeV/n 56-Fe ions. Data are already available for gamma irradiated animals: mice exposed to radiation developed multiple colonic lesions, but not control animals. Furthermore, animals exposed to lower doses survive up to 9 months after exposure and the size of colonic tumors is significantly larger than in 5 Gy irradiated mice.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
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